

REA Group Asia: Using data analysis to gain unique insights and enhance operating efficiencies

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

[About REA Group Asia](#)



With Google Cloud, REA Group Asia has the ability to capture and consolidate unstructured, semi-structured, and structured data in a single source, democratize data access and enable business to provide

About REA Group Asia

REA Group Asia connects property buyers and renters to agents or property vendors such as developers or individual owners. In Asia, REA Group operates in six countries and runs eight brands across the region.

Industries: Real Estate & Construction

Location: Malaysia

[Cloud Pub/Sub](#)

relevant experience
users.

Google Cloud results

- Enables one person to dedicate minimal time to managing data infrastructure
- Democratizes data across the organization and its eight brands across six countries
- Minimizes cost of data queries

Managed analytics report of data

Google Cloud (<https://cloud.google.com/>) ud
functions

BigQuery (<https://cloud.google.com/bigquery/>)

Firestore (<https://cloud.google.com/firestore/>)

Cloud Composer

(<https://cloud.google.com/composer/>)

Data Studio

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Analytics 360

(<https://marketingplatform.google.com/about/analytics-360/>)

REA Group (<https://www.rea-group.com/>) has come a

long way from its beginnings in a garage in

Melbourne's eastern suburbs. The business is now a

multinational operation that operates residential and

commercial property websites in Australia, a

property website in China, investments in property websites in India and North America, and a mortgage broking franchise and property data services provider. In Asia, REA Group owns a number of property portals, operates in six countries, and runs eight brands throughout the region.

“Data is so important to drive and support our strategic pillars, which are creating and delivering relevant, trusted content for our users, providing a high-quality experience to users that visit our website or use our app, and supporting our customers by

providing great
business value.”

*—MD Alauddin, Regional
Head of Data Science and
Engineering, REA Group
Asia*

REA Group Asia provides a platform—accessible via the web or mobile app—that connects property buyers and renters to agents or property vendors such as developers or individual owners. The REA Group Asia data team reports to MD Alauddin, Regional Head of Data Science and Engineering for Asia, and works with the regional technology team headed by Asia’s Chief Technology Officer, Shen Tham. “Our data team supports product development and provides data acquisition and processing, storage, management and governance services,” says Alauddin. “We also create business intelligence dashboards to support our operations and run data analytics and extract, transform, and load activities.”

Data key to success

Unsurprisingly for an established multinational digital advertising provider, using data to create value is a key success factor. “Data is so important to drive and support our strategic pillars, which are creating and delivering relevant, trusted content for our users, providing a high-quality experience to users that visit our website or use our app, and

supporting our customers by providing great business value,” explains Alauddin.

However, REA Group Asia was experiencing problems deduplicating its location data, consolidating it to a single repository, and making the relevant views accessible to business stakeholders. “Lack of a single source made it difficult for stakeholders to connect data, collaborate, and obtain actionable insights,” says Alauddin. “To deliver trusted content, we needed to ensure data quality at every stage of the process. Further, we had to understand why users were visiting our website or using our app, how long they used the website or app, and why they finished a session when they did.”

Through the use of [Google Analytics 360](https://marketingplatform.google.com/about/analytics-360) (https://marketingplatform.google.com/about/analytics-360)

, REA Group Asia gained unique insights about how consumers were experiencing its products and services and then tailored those experiences accordingly.

“For example, we wanted to understand how consumers engaged with individual property listings so we could provide relevant services to customers who invest in one or more packages that we offer. This would enable us to run simulations based on historical data to proactively advise customers how they can change listings or prices to win more business,” says Alauddin.

“We also wanted to help our internal teams with analyses that could help them achieve business goals such as increasing lead generation and reducing digital marketing cost per acquisition.”



MD Alauddin, Regional Head of Data Science and Engineering, REA Group Asia

Becoming data driven

Ultimately, the data team wanted to help the business move from making decisions based on

perception and instinct to becoming data driven.

“We are trying to overlay every decision we make with relevant data to determine whether it will deliver the most optimal result,” says Alauddin.

When Alauddin arrived at REA Group Asia in mid-2018, the data team was already loading Google Analytics 360 data into a BigQuery data warehouse to obtain insights to support decision-making.

However, the business was capturing other data and simply storing it in a range of sources, including legacy databases. Alauddin quickly determined REA Group Asia needed to adopt a more strategic approach, including consolidating data to a single location.

“Our first challenge was to implement the right data platform to manage 1 million+ records received every day from multiple sources,” says Alauddin.

“Much of this was unstructured data from property listings, transactions, weblogs, digital marketing, and JSON, typically data sent from a server to a web page. We needed to ingest this data as fast as we could, apply some standardization, and provide regional views that allowed product owners in different markets across Asia to use data in accordance with local business needs. We also had to process this data as fast as we could to serve owners and teams with the insights they need to make timely decisions.

“This meant building a platform that could integrate several systems and support a large number of workflows; manage data acquisition, cataloging and storage; accommodate queries; conduct staging and

cleansing; and create clean data marts that only authorized stakeholders could access.”

BigQuery ease of use

The business evaluated [BigQuery](#) (/bigquery) against competing products and found a number of superior features and functions, including the analytics data warehouse being much easier to use. “With BigQuery, we did not really have to concern ourselves with touchpoints, integrations or cataloguing—we simply needed to load the data for analysis,” says Alauddin.

The REA Group Asia data team then worked closely with the Google Cloud engineering teams in Malaysia and Singapore to develop a data platform that would support the Group’s roadmap. The engagement included daily workshops and service and architecture recommendations covering BigQuery and the data acquisition process.

As well as BigQuery, REA Group Asia’s data architecture now comprises the [Cloud Composer](#) (/composer) workflow orchestration service built on Apache Airflow (which enables REA Group Asia to author 400+ data pipelines), [Dialogflow](#) (/dialogflow) for data transformation and enrichment, [Pub/Sub](#) (/pubsub) for event ingestion and delivery to support stream analytics, and [Cloud Functions](#) (/functions) to build and connect its cloud services. [Filestore](#) (/filestore) provides a scalable database for mobile and web development, while [Looker Studio](#)

(<https://marketingplatform.google.com/about/data-studio/>)

delivers data visualization and reporting.

“Our key objectives are to iterate products more quickly and make faster decisions. Now we can connect multiple data sources, build use cases, and deliver to production quickly, enabling faster product activation and automating, which is key to the success of digital marketing.”

—MD Alauddin, Regional
Head of Data Science and

Providing open access to data

REA Group Asia's data strategy now provides open access to data across the business. In 2018, the team was using other data tools that were expensive and only provided a fixed number of licenses.

"Previously we couldn't provide access to all the team members who wanted to view and analyze data in their own way. They had to send a request to the analytics team for scheduling, which created bottlenecks and dissuaded teams from using analytics. We can now provide teams such as digital marketing with region- or country-specific views so they can generate reports and analysis themselves, improving overall operating effectiveness and the experience we deliver to customers and consumers," says Alauddin.

In addition, REA Group Asia can now warehouse and query data much more cost effectively. Not only was BigQuery much cheaper than alternative solutions, its ease of use meant there was a reduction in employee time required to manage the Group's data infrastructure.

“Combine this
with enabling
team members
to create use
cases and
business
intelligence
through a self-
service
approach using
BigQuery, and
our business is
more agile,
responsive, and
data driven—
which is our
ultimate goal.”

*—MD Alauddin, Regional
Head of Data Science and
Engineering, REA Group
Asia*

The qualities of the new data platform came to the fore when Alauddin’s GA-tracking implementation team delivered a new tracking service that relied on the rapid processing of a large volume of structured,

semi-structured, and unstructured data. “Data from any source is now easy to ingest and process, so launch of the new service was seamless,” says Alauddin.

“Our key objectives are to iterate products more quickly and make faster decisions. Now we can connect multiple data sources to provide insights and intelligence to build business cases more quickly, enabling faster product activation and automating, which is key to the success of digital marketing.

“Combine this with enabling team members to create use cases and business intelligence through a self-service approach using BigQuery, and our business is more agile, responsive, and data driven—which is our ultimate goal.”

Long-term growth opportunities

REA Group Asia remains poised for ongoing growth and to provide an even greater competitive challenge to other online property services across the region. The organization’s chief executive, Henry Ruiz, recently told the [Financial Review](https://www.afr.com/property/residential/rea-group-warns-propertyguru-we-re-coming-for-you-20190913-p52r1p) (https://www.afr.com/property/residential/rea-group-warns-propertyguru-we-re-coming-for-you-20190913-p52r1p)

, “We have our sights set on having a big impact across Asia.” The business aims to focus on professional agents that will provide a great service to property seekers who will use its digital assets. “We continue to improve and enhance our products

and services, and the business models that sit beneath them, to drive growth," says Ruiz.

**Learn
More**

Reach out to
our team to
see how
Google Cloud
can help your

business.

